

PSOP

Physical Security Observability Platform

Product Specification, Competitive Benchmark and Implementation Backlog

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Source base: live Domotz laboratory review, SecuriThings visual benchmark, PSOP product decisions and existing laboratory constraints.

Product thesis PSOP must not become a generic network-monitoring clone. Its defensible advantage is evidence-driven physical-security observability: proving what is known, how it was observed, how recent it is, and what physical operation is affected.

Status: Authoritative working baseline for architecture, prioritization and implementation.

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How to use this document Sections 1-8 define product intent and architecture. Sections 9-11 control execution. Section 12 preserves evidence and prevents assumptions from being mistaken for verified facts.

1. Executive decision

The benchmark validates that enterprise operators expect inventory, topology, alerting, dashboards, remote access, reporting, integrations, profiles and role-based access. PSOP should provide these capabilities selectively, but organize them around physical-security outcomes rather than network objects alone.

Decision	Position
Product category	Physical-security observability and operational evidence; not a general-purpose RMM.
Primary object	Physical asset and its operational role, with network interfaces and observations modeled separately.
Primary truth	Time-stamped evidence with provenance, confidence, freshness and conflicts.
Primary outcome	Detect and explain lost surveillance, recording, access-control or dependency capability.
Near-term proof	Speco N8NRL + Hikvision cameras + TP-Link lab, with the gateway started manually for tests.

2. Non-negotiable product principles

Principle	Mandatory interpretation
Evidence before status	Online is not equivalent to operational. Every material assertion must identify its evidence and age.
Observed is not declared	Manual metadata must never overwrite or masquerade as discovered/verified values.
Asset is not interface	A physical device may expose multiple interfaces, addresses, endpoints and conflicting identities.
Incident is not alert	Signals must be correlated, deduplicated and converted into operational incidents with impact.
Dependency-aware	Camera, PoE, switch, gateway, recorder, storage and WAN failures must be modeled as dependencies.
Safe automation	Read, propose and execute are distinct privilege levels; destructive actions require explicit controls.
Manual gateway	The PSOP gateway must not auto-start silently on the user's Mac; tests are manually initiated and ended.

3. Competitive benchmark findings

3.1 Inventory and asset experience

- Domotz provides Organizations, Collectors and Devices with global counters, search, filters, export and configurable columns.
- Inventory fields include collector, device name, status, type, make/model, IP/MAC, serial number, organization, location/zone, importance, tags, credentials, sensors, alert rules and first seen.
- The device drawer exposes Info, Connect, Alerts, History, SNMP and TCP plus connectivity and latency data.
- A lab object named Paulo's MacBook Pro combined Bonjour identity, hostname gateway, VMware OUI and Media Player type, demonstrating identity conflict and the danger of collapsing signals into one record.

PSOP consequence Create an Asset Intelligence drawer with Overview, Evidence, Capabilities, History, Incidents, Dependencies and Actions. Model physical asset, network interface, endpoint and observation independently.

3.2 Connectivity, security and remote access

- Connectivity includes uptime, outages, network information, route analysis and manual/automatic speed tests.
- Security includes new/rejoined-device events, TCP port scanning and UPnP forwarding checks.
- Remote access includes HTTP/HTTPS, custom connections and TCP tunnels with traffic accounting.
- Connectivity evidence does not prove video capture, recording, storage health or operational coverage.

3.3 Alerts, dashboards, profiles and reporting

- Alert activity supports timeline and filters; rules include device down, IP conflict, TCP service, new device and network-setting changes.
- Dashboards use ready-made monitoring tables and manufacturer/purpose templates.
- Profiles bulk-apply alerts, information and SNMP sensors using filters and schedules, with a 30-day application history.
- Reports can be manual or scheduled and include connectivity, WAN performance and inventory composition.

3.4 Integrations, drivers and AI

- Software integrations include documentation, PSA and ticketing platforms; messaging includes email, push, Slack, Teams and Telegram.
- The driver catalog is extensible: catalog drivers can be copied and edited, and custom drivers can expose tables and actions.
- No Hikvision-specific driver was observed. The ONVIF driver was tested on LILIN LD2222.

- AI Agents expose one MCP endpoint to in-app chat, ChatGPT, Claude, Copilot Studio and generic clients.
- The in-app agent displayed tool calls and arguments, but incorrectly inferred good standing from zero triggered alerts.

3.5 ONVIF driver technical assessment

The reviewed driver uses ONVIF GetProfiles and GetStreamUri, then sends RTSP DESCRIBE and labels a profile OK when the response contains 200 OK. This verifies endpoint negotiation, not actual visual capture or recording.

Observed behavior	Assessment / risk
GetProfiles	Useful profile discovery; empty results must not validate successfully.
GetStreamUri	Useful RTSP discovery; visible code appears to reference profile.profiletoken instead of profileToken.
RTSP DESCRIBE	Proves an endpoint can describe a stream, not that frames are current, usable or recorded.
Fixed port 554	May fail on nonstandard RTSP deployments or URIs with explicit ports.
Array used as URI	Visible code uses stream.uris at critical points instead of the current uri.
Error callback	Some error paths appear to log without resolving/rejecting, risking timeout/hanging behavior.
30-minute minimum sample	Too slow for critical camera-loss detection unless complemented by faster probes.

4. Adopt / adapt / surpass / defer decisions

Capability	Decision	Rationale
Inventory table + side drawer	ADOPT	Core operator pattern; enrich with evidence and conflicts.
Organizations / collectors	ADAPT	Use Tenant/Customer → Site → Building/Area → Gateway/Recorder → Device.
Network topology	SURPASS	Offer Physical, Observability and Dependency relationship views.
Device online status	SURPASS	Split reachability, device health, video, recording, storage and dependency states.

Capability	Decision	Rationale
Monitoring profiles	ADAPT	Versioned Monitoring Policy Profiles with staged rollout and rollback evidence.
Alert rules	ADAPT	Add persistence, window, deduplication, dependency and physical impact.
Remote access	DEFER	High risk; add only after RBAC, audit, expiry and incident linkage.
Custom code editor	DEFER	Start with signed/versioned adapters; free-form scripting raises security complexity.
MCP multi-client layer	ADOPT LATER	Build provider-neutral interface after the evidence model is reliable.
Executive dashboard	ADAPT	Use SecuriThings-like hierarchy but show evidence coverage and physical impact.

5. Target product architecture

Layer	Responsibility
Experience	Executive Overview; Fleet Workspace; Asset Intelligence; Topology; Incident Rail; Reports; Administration.
Operational domain	Assets, interfaces, endpoints, capabilities, dependencies, incidents, policies and maintenance.
Evidence engine	Observations, provenance, confidence, freshness, conflicts, assertions and evidence chains.
Detection & correlation	Rules, temporal windows, deduplication, suppression, dependency/root-cause and impact.
Collection	Manual-start gateway, ONVIF, RTSP, HTTP/API, SNMP, ICMP/TCP, NVR/vendor adapters.
Integration	Notifications, tickets, reports, webhooks and later provider-neutral MCP tools.
Governance	Tenant isolation, RBAC/ABAC, secrets, audit, retention, consent and destructive-action controls.

5.1 Core domain model

Entity	Purpose	Critical relationships
PhysicalAsset	Stable real-world equipment identity.	Belongs to site/area; owns interfaces; fulfills capabilities.
NetworkInterface	MAC/IP/VLAN and address history.	Belongs to asset; exposes endpoints; produces observations.
Endpoint	ONVIF, RTSP, HTTP, SNMP or vendor API surface.	Belongs to interface; tested by probes.
Observation	Immutable time-stamped measured fact.	Source, collector, method, raw value, time and quality.
Assertion	Current interpreted statement about an asset.	Derived from evidence; confidence; valid/fresh interval; conflicts.
Capability	Expected function such as LIVE_VIDEO or RECORDING.	Depends on assets/endpoints and is evaluated by evidence.
Dependency	Physical/observability/operational relationship.	Connects camera, PoE, switch, gateway, NVR and storage.
Incident	Correlated operational problem.	Impact, evidence chain, affected capabilities, lifecycle and ownership.

5.2 Required state dimensions

Dimension	Examples	Must not be collapsed into
Reachability	ICMP/TCP/HTTP reachable	Operational
Device health	Uptime, temperature, firmware	Video
Video health	Frames, freeze, black image, bitrate	Recording
Recording health	Continuous write, gaps, retrieval test	Storage capacity
Storage health	Disk state, utilization, retention	NVR reachability
Evidence health	Freshness, coverage, confidence	No active alerts

6. Operational Evidence Foundation

This is the first implementation increment because every dashboard, alert, incident and AI answer depends on trustworthy observations.

6.1 Evidence levels for video assurance

Level	Assertion	Minimum proof
E0	Unknown	No recent observation.
E1	Endpoint reachable	Network service responds.
E2	ONVIF profile discovered	GetProfiles returns a valid profile/token.
E3	RTSP URI obtained	GetStreamUri succeeds for that profile.
E4	RTSP session negotiated	DESCRIBE/SETUP/PLAY succeeds as applicable.
E5	Frames received	Decoded or sampled frames are current.
E6	Visual quality validated	Freeze/black/obstruction/quality checks evaluated.
E7	Recording proven	NVR reports recording and timeline continuity.
E8	Retrieval proven	A recent recorded segment can be requested and validated.

6.2 Provenance and confidence

- VERIFIED: corroborated by a high-authority source or multiple independent observations.
- OBSERVED: directly measured but not necessarily corroborated.
- DECLARED: manually supplied by a user or imported from administrative data.
- INFERRED: derived by rules or models from other evidence.
- NOT_VERIFIED: value exists but lacks sufficient current evidence.

Every material assertion must expose source, observed_at, received_at, valid_until, confidence, method, collector/gateway, raw evidence reference and conflicts.

6.3 Required timestamps

Timestamp	Meaning
First Seen	Earliest reliable observation of the identity/interface.
Last Seen	Most recent observation of any kind.
Last Verified Identity	Most recent corroborated identity evidence.
Last Operational Evidence	Most recent evidence proving the relevant capability.
Evidence Age	Current time minus last operational evidence, evaluated against policy.

7. Executive Operations Overview

The SecuriThings visual benchmark is useful because it separates sites, devices, management systems, alerted devices, total alerts, maintenance, compliance and geography. PSOP should retain this hierarchy while replacing generic status with operational proof.

Dashboard region	PSOP content	Operator question
Top summary	Sites, monitored assets, gateways/recorders, affected capabilities	What is the operational footprint and what is currently affected?
Fleet composition	Cameras, NVRs, access control, network/support assets	What equipment is under responsibility?
Situation center	Affected assets, active incidents, severity and business/physical impact	What requires action now?
Evidence coverage	Fresh, stale, missing and conflicting operational evidence	Can the platform substantiate its status?
Maintenance	Firmware, credentials, storage, lifecycle, certificates	What degradation is predictable or preventable?
Compliance	Policy coverage with numerator, denominator and evidence timestamp	What exactly does each percentage measure?
Map	Site state and affected areas with drill-down	Where is the risk concentrated?

Design direction Keep the benchmark's airy executive hierarchy, but use the PSOP Shield Wave identity, controlled depth and operational colors. Avoid pure black and avoid decorative effects that reduce scanability.

8. Access, AI and integration security

8.1 Authorization model

Domotz demonstrates Users → Groups → Roles, module permissions and collector scope. PSOP should extend this to action + resource + scope + condition.

Role	Default intent
Platform Administrator	Platform governance; no implicit access to tenant secrets without audited elevation.

Role	Default intent
Tenant Administrator	Users, sites, policies and integrations within one tenant.
Security Operations Manager	Incident operations, policies and reporting.
Operator	Monitor, triage, annotate and escalate.
Field Technician	Assigned assets/sites, diagnostics and approved maintenance actions.
Incident Responder	Time-bound access linked to assigned incidents.
Auditor / Read-Only	Evidence, history and audit visibility without mutation.
Integration Service Account	Narrow machine-to-machine scope, expiry and rotation.
AI Agent	Explicit tool scopes; read/propose/execute separation.

8.2 AI Operations Assistant

Mode	Allowed behavior	Control
READ	Retrieve assets, evidence, incidents and policies.	Tenant/site/tool scopes; tool trace.
PROPOSE	Recommend actions with reason, expected impact and evidence.	No mutation; human approval path.
EXECUTE	Run an approved diagnostic or operational action.	Step-up auth, confirmation, expiry, audit, rollback where possible.
PROHIBITED	Silent remote access, secret retrieval, destructive changes without authority.	Hard policy denial and logged attempt.

- Every answer must distinguish tool result from model interpretation.
- Zero alerts must never be translated automatically into healthy operation.
- Trace must record tool, arguments, result, errors/retries, duration, user/agent identity, scope, permission and evidence links.
- OAuth and short-lived scoped credentials are preferred; long-lived bearer tokens are restricted to controlled automation.

9. Prioritized implementation backlog

Priority	Domain	Backlog item	Definition of value
P0	Evidence	PhysicalAsset / NetworkInterface / Endpoint split	Prevents identity collapse and supports conflicts.

Priority	Domain	Backlog item	Definition of value
P0	Evidence	Immutable Observation model	Stores source, timestamps, method, raw reference and quality.
P0	Evidence	Assertion + confidence + freshness	Exposes VERIFIED/OBSERVED/DECLARED/INFERRED/NOT_VERIFIED.
P0	Evidence	Identity conflict detection	Shows contradictory hostname, OUI, vendor, type and manual values.
P0	Collection	Gateway manual execution lifecycle	Start/stop/status; never silent background start.
P0	Collection	ONVIF discovery	Profiles, tokens, device/service information and error classification.
P0	Collection	RTSP evidence probes	URI discovery, negotiation, frame freshness and structured failures.
P0	Policy	Evidence freshness policies	Expected cadence and maximum age by asset/capability profile.
P0	UI	Asset Intelligence drawer	Overview, Evidence, Capabilities, History, Incidents, Dependencies, Actions.
P0	Incidents	Signal-to-incident pipeline	Persistence, windows, deduplication and lifecycle.
P1	Topology	Operational Topology	Physical, Observability and Dependency views.
P1	Collection	NVR/Speco recording evidence	Channels, recording state, storage and retrieval proof.
P1	Collection	Hikvision adapter	Identity, firmware, stream and device telemetry with lab evidence.
P1	Dashboard	Executive Operations Overview	Operational footprint, impact, evidence coverage and map.
P1	Policy	Monitoring Policy Profiles	Versioned policy, filters, staged application, result and rollback.
P1	Notifications	Incident routing and escalation	Role/site/severity schedule, delivery confirmation and escalation.
P1	RBAC	Action-resource-scope-condition authorization	Tenant isolation and least privilege.

Priority	Domain	Backlog item	Definition of value
P1	Audit	Unified audit trail	Human, service and agent actions with immutable history.
P2	Reports	Scheduled operational reports	Weekly/monthly/custom, recipient profiles and comparisons.
P2	Integrations	Webhooks and ticketing	Idempotent incident synchronization and delivery status.
P2	AI	Read-only MCP tools	Evidence-grounded provider-neutral queries.
P3	AI	Propose and execute modes	Controlled actions after RBAC and audit maturity.
P3	Remote	Contextual remote access	Incident-linked, expiring, audited and never silently enabled.
P3	Extensibility	Signed adapter SDK	Versioned official adapters before unrestricted scripting.

10. Acceptance criteria

Area	Acceptance criterion
Identity	A conflicting OUI, hostname and manual vendor are displayed as separate claims with source and confidence; no silent overwrite.
Freshness	The UI shows last operational evidence and computes stale/missing according to the assigned policy.
Video	A reachable camera without recent frames cannot be labeled operational.
Recording	An online NVR without verified recording continuity produces a distinct unknown/degraded state.
Incident	Repeated related observations create one incident, preserve the signal timeline and expose affected capabilities.
Dependencies	A parent PoE/NVR/gateway failure correlates downstream symptoms and identifies probable root cause without hiding evidence.
Dashboard	Every percentage shows numerator, denominator, scope and evidence timestamp.

Area	Acceptance criterion
Authorization	A user outside the permitted tenant/site cannot query, view, mutate or infer protected asset data.
AI	The assistant cites evidence, shows tool traces and refuses unauthorized actions; zero alerts is not reported as proven health.
Gateway	The gateway remains stopped after reboot unless the user explicitly starts it for a test.

11. Delivery roadmap

Phase	Outcome	Scope
Phase 0	Repository audit and baseline	Map current frontend/backend/schema/tests to this specification; preserve working incident-notification work.
Phase 1	Operational Evidence Foundation	Domain migrations, evidence services, conflict/freshness logic, gateway lifecycle and tests.
Phase 2	Lab proof	Hikvision/ONVIF/RTSP collection and Speco NVR evidence, with controlled failure scenarios.
Phase 3	Operator experience	Asset Intelligence drawer, evidence timeline, capabilities and operational topology.
Phase 4	Incident intelligence	Correlation, dependency analysis, impact, ownership, routing and escalations.
Phase 5	Executive experience	PSOP Executive Operations Overview, maintenance, compliance and reports.
Phase 6	Governed integrations and AI	RBAC/audit completion, webhooks/ticketing, read-only MCP, then propose/execute.

Delivery discipline Each phase must ship schema, API, UI, tests, observability and migration/rollback notes together. A visually complete screen backed by synthetic or ambiguous status is not considered complete.

12. Evidence register and unresolved questions

12.1 Evidence sources

Source	Coverage	Use in this specification
Domotz live trial, 4-5 Sep 2026	Collector, inventory, device drawer, dashboards, alerts, profiles, integrations, drivers, AI Agents, RBAC.	Observed product behavior and limitations.
ONVIF driver source text	Custom JavaScript driver for profile discovery and RTSP DESCRIBE.	Technical assessment of proof depth and apparent defects.
SecuriThings dashboard image	Sites/devices/systems, alerts, maintenance, compliance and map.	Executive dashboard composition reference only.
PSOP laboratory constraints	Speco N8NRL, three Hikvision PoE cameras, TP-Link AX3000 and Lorex legacy system.	Implementation validation environment.
Existing PSOP decisions	Hybrid UI, Fleet Topology, Incident Rail, manual gateway and notification work.	Continuity and constraints for repository audit.

12.2 Unresolved questions to verify in the repository/lab

- Which current database entities already represent device, site, gateway, incident and observations?
- Which notification/escalation changes on feat/incident-notifications-escalation are complete and tested?
- What ONVIF data is already collected from the Hikvision cameras and through which interfaces?
- Which Speco N8NRL APIs, ONVIF services or vendor endpoints expose channel, recording, storage and firmware evidence?
- How are tenant isolation, secrets and audit currently implemented?
- Which existing UI components can be evolved without discarding the established PSOP visual identity?

12.3 Immediate next action

1. Open the PSOP repository in the working environment.
2. Run a non-mutating repository inventory and test baseline.
3. Create a gap matrix mapping current implementation to P0 backlog items.
4. Propose the smallest safe migration sequence for the Operational Evidence Foundation.
5. Implement and verify the first vertical slice using real lab evidence.

Final control This document is a product baseline, not proof that each capability already exists. Repository and laboratory evidence must confirm implementation status before any feature is represented as complete.